JOSEPH ALEXANDER GILSOUL

Layton, UT | 385-445-8702 | agilsoulphysics@gmail.com | https://github.com/AGilsoul/My-Projects

SKILLS/CERTIFICATIONS

C++, C#, Java, Python, F Prime, SQL, CFD, CAN, EMT-B, BLS

WORK/EXTRACURRICULAR EXPERIENCE

Software Engineer Intern
Optisys
01/2025-12/2025
Salt Lake City, UT

- Worked as the sole software engineer for six months, followed by six months with other interns.
- Automated mechanical and RF engineering design processes using Python in Siemens NX and Ansys HFSS.
- Set up our own local Git server where I managed versioning for our software.

Research Assistant
University of Utah Summer Program for Undergraduate Research (SPUR) and Undergraduate Research Opportunity Program (UROP)
05/2023-12/2023, 08/2024-10/2024
Salt Lake City, UT

- Modelling and Simulation of Turbulent Premixed Hydrogen Flames.
- Worked under Dr. Alex Novoselov as part of the Numerical Turbulence, Energy, and Reactions Group.
- Wrote codes to analyze the data generated from Direct Numerical Simulations (DNS) using CFD of Hydrogen flames and extract 1-dimensional flame paths following a gradient in 3-dimensions.

Research Assistant
University of Utah Department of Mathematics Research Experience for Undergraduates (REU)
08/2023-12/2023
Salt Lake City, UT

- Modelled the responses of resistant bacteria to antibiotics using stochastic differential equations.
- Implemented stochastic modelling algorithms such as the Gillespie Algorithm and Tau Leaping in Python.
- Mitigating Antibiotic Resistance From Missed Doses published in Journal of Pharmacokinetics & Pharmacodynamics

EDUCATION

Degree: B.S. Applied Mathematics, Minor Computer Science
University of Utah (Expected Graduation Date: December 2026)
08/2022-Present
GPA: 3.9
Relevant Coursework:

- Calculus III
- Applied Statistics I
- Software Practice II
- Algorithms
- Computer Systems

Degree: Programming Essentials Certificate of Proficiency, Early College Program
Weber State University
08/2020-04/2022
GPA: 3.99
Relevant Completed Coursework:

- Intro to Database Design & SQL
- Advanced Algorithms